



FADO

Meeting 2

Tiago Silva

CISUC · Data-Centric AI Lab

Department of Informatics Engineering, University of Coimbra



Outline

1. **Weekly reviews:** KANs;
2. **Review pass:** what we found, what we fixed, a correction;
3. **Decisions taken;**
4. **The λ controller:** the answer to the novelty critique;
5. **First results** and how they happened;
6. **Ablation plan**, open questions and next steps.

KANs

KAN: Kolmogorov–Arnold Networks

Ziming Liu^{1,4*} Yixuan Wang² Sachin Vaidya¹ Fabian Ruehle^{3,4}
James Halverson^{3,4} Marin Soljačić^{1,4} Thomas Y. Hou² Max Tegmark^{1,4}

¹ Massachusetts Institute of Technology

² California Institute of Technology

³ Northeastern University

⁴ The NSF Institute for Artificial Intelligence and Fundamental Interactions

Kolmogorov-Arnold Networks

Activation functions are placed on edges instead of nodes.

D^3 KAN: Dual Denoising and Data-Complexity-Adaptive Kolmogorov-Arnold Networks for Long-Term Time Series Forecasting

Minrui Wang^{1,2}, Jiwei Qin^{1,2*}, Di Wang^{3,4}, Dacheng Wang⁵,
Zhiqi Shen^{3,4}

D^3 KAN

A denoising and data-complexity-adaptive KAN for long-term forecasting.

The Fortieth AAAI Conference on Artificial Intelligence (AAAI-26)

Learning Fair Representations with Kolmogorov-Arnold Networks

Amisha Priyadarshini¹, Sergio Gago-Masague¹

¹Department of Computer Science, University of California, Irvine, USA
apriyad1@uci.edu, sgagomas@uci.edu

Learning Fair Representations with KANs

KANs applied to learning fair representations.

The review pass: results we could not trust

Invalid results

COMPAS loader cut every split to **10 rows**: every stored COMPAS number came from 10 test samples.

Significance script turned untestable comparisons ($n < 2$) into $p = 0$, ***.

Silently inert experiments

COMPAS defaults overwrote the swept flags: every cell of the old sensitivity sweep ran **one** configuration.

The W&B summary crashed before writing the metric the sweeps optimise.

Arms not comparable

On COMPAS, FADO/Base used a 250-sample window, ARF/RFR 1000.

Pre-training used $\lambda = 0.1$, evaluation $\lambda = 1.0$.

All fixed; COMPAS must be **re-run**.

Checked and fine: no stored run hit the variable-dispatch collision ($d = 108, 19, 760$ vs 2^{depth}).

A correction to Meeting 1

“**37.12 → 13.22 minutes on Adult**” timed a benchmark copy of the recursive updater, **not the code the pipeline ran**.

The production path is now the level-wise vectorised updater, chosen by depth (`'auto'`).

The benchmark will be rewritten to time the production forest, and the headline number replaced.

The complexity argument, $O(B \times 4^n) \rightarrow O(B \times 2^n)$, stands; the wall-clock figure does not until re-measured.

Decisions taken

Scope	Exclusively online learning.
λ comparison	Report accuracy-vs-DP trade-off curves over one shared λ grid; no λ is picked.
Tuning budget	Controller ablations vary only the controller; parameters that reach every arm (λ , depth, trees, windows) are swept for every arm.
Metrics	Whole-stream and per-phase / post-drift accuracy and DP.
Search	Grid for factorials, random for sensitivity; report distributions, never an argmax.
Seeds	Tune on 11–55, report on 66–202. COMPAS splits by seed only (two scenarios).
Per-tree λ	Deferred; at most an ablation.

Answering novelty: λ becomes a dual variable

Constraint

DP = $\max_g |D_g|$, with D_g the gap between group g and the mean positive rate.

Target: $+D_g \leq \varepsilon$ and $-D_g \leq \varepsilon$ for every g .

Signal

One sample gives $v_t[g]$ with $\mathbb{E}[v_t[g]] = D_g$ (Horvitz–Thompson).

Unbiased, no window, no autocorrelation.

Update, every sample

$$\mu \leftarrow \text{clip}(\mu + \eta(\pm v_t[g] - \varepsilon), 0, \lambda_{\max})$$
$$\lambda_t = \min(\lambda_{\text{base}} + \sum \mu, \lambda_{\max})$$

Plus forgetting: every confirmed drift resets the penalty's statistics, so λ scales a *post*-drift gradient.

Why not feed the rolling DP?

Stationary 50k stream: every alarm is false.

signal	lag-1 autocorr.	alarms, $\delta=10^{-5}$
per-sample estimate	-0.001	0
rolling DP, $W=250$	0.991	24
DP every W samples	-0.044	0

Detectors assume independent inputs; a rolling DP shares $W-1$ of W samples with its predecessor.

The per-sample estimate keeps both the detector's bound and an **unbiased** constraint gradient.

Limits: soft trees are non-convex, so primal-dual bounds are motivation, not a guarantee.

First results: DP held down after the drift

COMPAS abrupt_race, 3 tuning seeds, $\lambda_{\text{base}} = 0.3$, $\varepsilon = 0.05$.

arm	accuracy	DP	post-drift DP
FADO (λ controller)	0.605	0.058	0.066
FADO, no reset	0.606	0.081	0.099
FADO, λ fixed	0.605	0.094	0.117
Aranyani-Base	0.617	0.093	0.116

Same direction on every seed: -54% , -18% , -40% DP vs. λ fixed.

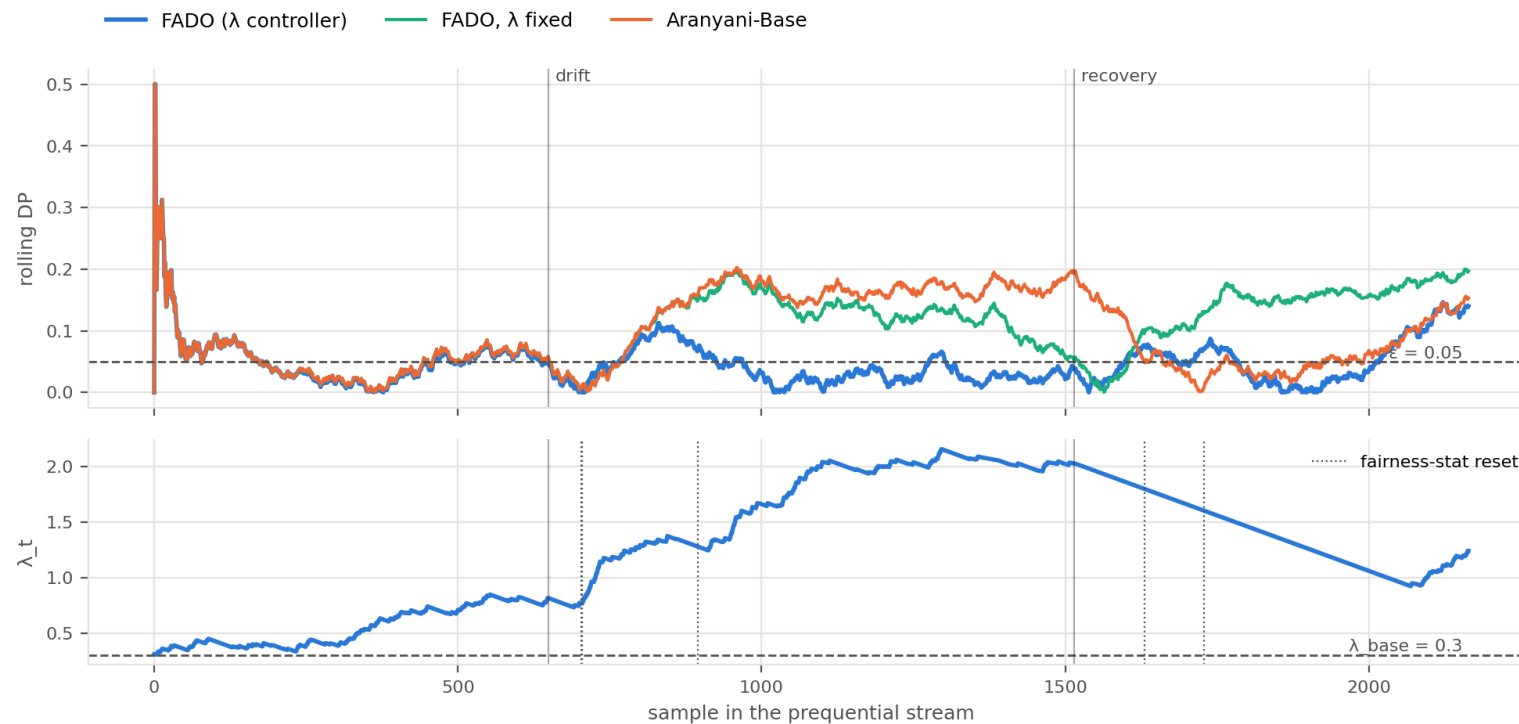
No accuracy cost: $\sim 12\%$ of predictions change; half become right, half wrong. Borderline cases move.

Not a collapse: positive rate $0.40 \rightarrow 0.37$.

A fair classifier still shows rolling DP ≈ 0.026 (window noise). 3 seeds, one scenario: a direction, not a claim.

How it happened: the price rises while DP is above target

compas · abrupt_race · seed 11



Identical until the drift; then the fixed- λ arms sit at DP 0.15–0.20 while λ climbs to 2 and holds FADO near ε .

Ablation plan: nine sweeps, ready to run

sweep	question	runs
component (COMPAS / Adult)	Does each part contribute? 9 presets, incl. <i>no-λ</i> , <i>λ-only</i> , <i>no-reset</i>	18 / 27
monitor	Which signal and detector should watch fairness?	48
sensitivity	Basin or knife edge? Controller parameters, incl. ε , η	120
λ budget (COMPAS / Adult)	Trade-off curves, same λ grid for FADO and Base	14 / 21
reference (COMPAS / Adult)	ARF / RFR points on those curves	2 / 3
architecture	Does the gap survive other depths and tree counts?	15

5 tuning seeds per run; every config is validated before launch.

Open questions

- ▶ **Target ε :** chosen per dataset on tuning seeds, or stated as a requirement?
- ▶ **Claim:** “after drift, FADO spends fewer samples above ε than Base”?
- ▶ **Theory:** a stability argument for non-convex soft trees, or state the convex-only caveat?
- ▶ **Tuning protocol:** disjoint streams, or prefix tuning (drift starts at 15–30%, so a prefix sees none)?
- ▶ **Accuracy gap to Base:** comes from the LR/temperature reaction, not from λ .

Next steps

1. Re-run COMPAS on full splits;
2. Launch the sweeps on the tuning seeds;
3. Metric for the claim: samples above ε after each change point;
4. Re-benchmark the production forest;
5. Final protocol: 20 reporting seeds, all datasets, all arms.